

| ID   | Name                                    | Direction       | Response | Access | Conformance  |
|------|-----------------------------------------|-----------------|----------|--------|--------------|
| 0x47 | <b>StopMoveStep</b>                     | client ⇒ server | Y        | O      | HS   XY   CT |
| 0x4B | <b>MoveCol-<br/>orTempera-<br/>ture</b> | client ⇒ server | Y        | O      | CT           |
| 0x4C | <b>StepCol-<br/>orTempera-<br/>ture</b> | client ⇒ server | Y        | O      | CT           |

### 3.2.8.1. Generic Usage Notes

When asked to change color via one of these commands, the implementation SHALL select a color, within the limits of the hardware of the device, which is as close as possible to that requested. The determination as to the true representations of color is out of the scope of this specification. However, as long as the color data fields of the received command are within the permitted range of this specification and no error condition applies, the resulting status code SHALL be SUCCESS.

For example the MoveToColorTemperature command: if the target color temperature is not achievable by the hardware then the color temperature SHALL be clipped at the physical minimum or maximum achievable (depending on the direction of the color temperature transition) when the device reaches that color temperature (which MAY be before the requested transition time).

If a color loop is active (i.e., the ColorLoopActive attribute is equal to 1), it SHALL only be stopped by sending a specific ColorLoopSet command frame with a request to deactivate the color loop (i.e., the color loop SHALL NOT be stopped on receipt of another command which has a 'stop' semantic such as the EnhancedMoveToHue command with MoveMode==Stop, or the StopMoveStep command). In addition, while a color loop is active, a manufacturer MAY choose to ignore incoming color commands which affect a change in hue.

### 3.2.8.2. Note on Change of ColorMode

The first action taken when any one of these commands is received is to change the ColorMode attribute to the appropriate value for the command (see individual commands). Note that, when moving from one color mode to another (e.g., CurrentX/CurrentY to CurrentHue/CurrentSaturation), the starting color for the command is formed by calculating the values of the new attributes (in this case CurrentHue, CurrentSaturation) from those of the old attributes (in this case CurrentX and CurrentY).

When moving from a mode to another mode that has a more restricted color range (e.g., CurrentX/CurrentY to CurrentHue/CurrentSaturation, or CurrentHue/CurrentSaturation to ColorTemperatureMireds) it is possible for the current color value to have no equivalent in the new mode. The behavior in such cases is manufacturer dependent, and therefore it is recommended to avoid color mode changes of this kind during use.

### 3.2.8.3. Use of the OptionsMask and OptionsOverride Fields

The OptionsMask and OptionsOverride Fields SHALL both be present. Default values are provided to interpret missing fields from legacy devices. A temporary Options bitmap SHALL be created from